Assignment 1, Combinatorics

1. Among 30 Students of the Informatics Management Class of 2025, how many ways are there to form a delegation of 7 members such that :

a) Student A is always included.

$$C(29,6) = \frac{n!}{r!(n-r)!} = \frac{29!}{6!(29-6)!} = \frac{29 \cdot \overset{7}{\cancel{28}} \cdot \overset{9}{\cancel{27}} \cdot \overset{13}{\cancel{26}} \cdot \overset{5}{\cancel{25}} \cdot \overset{4}{\cancel{24}} \cdot \cancel{23!}}{\cancel{6} \cdot \cancel{5} \cdot \cancel{4} \cdot \cancel{3} \cdot 2 \cdot 1 \cdot \cancel{23!}}$$

$= 29 \cdot 7 \cdot 9 \cdot 13 \cdot 5 \cdot 4$

$= 29 \cdot 63 \cdot 13 \cdot 20$

$= 1827 \cdot 260 = \underline{475.020}$ ways

b) Student A is not included.

$$C(29,7) = \frac{29!}{7!(29-7)!} = \frac{29 \cdot \overset{7}{\cancel{28}} \cdot \overset{9}{\cancel{27}} \cdot \overset{13}{\cancel{26}} \cdot \overset{5}{\cancel{25}} \cdot \overset{4}{\cancel{24}} \cdot 23 \cdot \cancel{22!}}{\cancel{7} \cdot \cancel{6} \cdot \cancel{5} \cdot \cancel{4} \cdot \cancel{3} \cdot \cancel{2} \cdot 1 \cdot \cancel{22!}}$$

$= 29 \cdot 9 \cdot 13 \cdot 5 \cdot 4 \cdot 23$

$= 261 \cdot 13 \cdot 20 \cdot 23$

$= 261 \cdot 260 \cdot 23 = \underline{\underline{1.560.780}}$ ways

c) Student A is always included, but Student B is not.

$$C(28,6) = \frac{28!}{6!(28-6)!} = \frac{\overset{7}{\cancel{28}} \cdot \overset{9}{\cancel{27}} \cdot \overset{13}{\cancel{26}} \cdot \overset{5}{\cancel{25}} \cdot \overset{4}{\cancel{24}} \cdot 23 \cdot \cancel{22!}}{\cancel{6} \cdot \cancel{5} \cdot \cancel{4} \cdot \cancel{3} \cdot \cancel{2} \cdot 1 \cdot \cancel{22!}}$$

$= 7 \cdot 9 \cdot 13 \cdot 5 \cdot 4 \cdot 23 = 819 \cdot 460$

$= \underline{\underline{376.740}}$ ways

d) Student B is always included, but Student A is not.

$$C(28,6) = \frac{28!}{6!(28-6)!} = \frac{\overset{7}{\cancel{28}} \cdot \overset{9}{\cancel{27}} \cdot \overset{13}{\cancel{26}} \cdot \overset{5}{\cancel{25}} \cdot \overset{4}{\cancel{24}} \cdot 23 \cdot \cancel{22!}}{\cancel{6} \cdot \cancel{5} \cdot \cancel{4} \cdot \cancel{3} \cdot \cancel{2} \cdot 1 \cdot \cancel{22!}}$$

$= 7 \cdot 9 \cdot 13 \cdot 5 \cdot 4 \cdot 23 = 819 \cdot 460$

$= \underline{\underline{376.740}}$ ways

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e) both A and B are included.

$C_{(28,5)} = \frac{28!}{5!(28-5)!} = \frac{\cancel{28}^{7} \cdot \cancel{27}^{9} \cdot \cancel{26}^{13} \cdot \cancel{25}^{5} \cdot 24 \cdot \cancel{23!}}{\cancel{5} \cdot \cancel{4} \cdot \cancel{3} \cdot \cancel{2} \cdot 1 \cdot \cancel{23!}}$

$= 7 \cdot 9 \cdot 13 \cdot 5 \cdot 24 = 98.280$ ways

F) at least one of the student named A or B is Include

A included, B not Included

$C_{(28,6)} = 376.790$

B included, A not Included

$C_{(28,6)} = 376.790$

Both A and B Included

$C_{(28,5)} = 98.280$

Add All Cases

$376.790 + 376.790 + 98.280$

$= 851.760$ ways

2. One day, a lecturer wants to choose 5 students to help with a project. There are 10 male students and 10 female students interested. If the requirements is that at least 3 members must be male, determine the number of ways the lecturer can choose the project members.

The answers :

There are 3 cases :

1. 3 males, 2 females

$C_{(10,3)} = \frac{10!}{3!(10-3)!} = \frac{10 \cdot \cancel{9}^{3} \cdot \cancel{8}^{4} \cdot \cancel{7!}}{\cancel{3} \cdot \cancel{2} \cdot 1 \cdot \cancel{7!}} = 10 \cdot 3 \cdot 4 = 120$

$C_{(10,2)} = \frac{10!}{2!(10-2)!} = \frac{\cancel{10}^{5} \cdot 9 \cdot \cancel{8!}}{\cancel{2} \cdot 1 \cdot \cancel{8!}} = 45$

then multiply $= 120 \cdot 45 = 5400$

2. 4 males, 1 female

$C_{(10,4)} = \frac{10!}{4!(10-4)!} = \frac{10 \cdot \cancel{9}^{3} \cdot \cancel{8} \cdot \cancel{7} \cdot \cancel{6!}}{\cancel{4} \cdot \cancel{3} \cdot \cancel{2} \cdot 1 \cdot \cancel{6!}} = 10 \cdot 3 \cdot 7 = 210$

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$C(10,1) = \dfrac{10!}{1(10-1)!} = \dfrac{10 \cdot \cancel{9!}}{\cancel{9!}} = 10$

then multiply $= 210 \cdot 10 = 2.100$

3. 5 males, 0 females

$C(10,5) = \dfrac{10!}{5!(10-5)!} = \dfrac{\cancel{10} \cdot 9 \cdot \cancel{8} \cdot 7 \cdot \cancel{6} \cdot \cancel{5!}}{\cancel{5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 \cdot 5!}} = 2 \cdot 9 \cdot 2 \cdot 7 = 252$

$C(10,0) = \dfrac{10!}{0(10-0)!} = \dfrac{10!}{10!} = 1$

then multiply $= 252 \cdot 1 = 252$

Add All cases:

$5.900 + 2.100 + 252 = \underline{\underline{7.752 \text{ ways}}}$

3. Determine the number of ways to permute the word SURABAYA. If it must begin with a consonant.

$S = \{S, U, R, A, B, A, Y, A\}$

letter S = 1
U = 1
R = 1
A = 3
B = 1
Y = 1

n = 8, but we must begin with a consonant means n = 7 and 4 distinct letters (appearing 1 time each)

number of ways $= 4 \cdot P(7; 3, 1, 1, 1, 1)$

$= 4 \cdot \dfrac{7!}{3!\,1!\,1!\,1!\,1!}$

$= 4 \cdot \dfrac{7 \cdot 6 \cdot 5 \cdot 4 \cdot \cancel{3!}}{\cancel{3!}}$

$= 4 \cdot 840$

$= 3360$ ways

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4. How many integer solutions are there to $x_1 + x_2 + x_3 + x_4 + x_5 = 27$ if $1 < x_1 \leq 5$, and $x_4 \geq 17$?

For $x_1 \geq 2$

For $x_4 \geq 17$

Total allocated $= 2 + 17 = 19$

$r =$ Total Sum $-$ Allocated

$= 27 - 19 = 8$

$n = 5$ (variables x_1 to x_5)

Using $C(n+r-1, r)$

$$C(5+8-1, 8) = C(12,8) = \frac{12!}{8!(12-8)!} = \frac{12!}{8!\,4!} = \frac{\overset{3}{\cancel{12}} \cdot 11 \cdot \overset{5}{\cancel{10}} \cdot \overset{3}{\cancel{9}} \cdot \cancel{8!}}{\cancel{8!}\ \cancel{4} \cdot \cancel{3} \cdot \cancel{2} \cdot 1}$$

$$= 3 \cdot 11 \cdot 5 \cdot 3 = 495$$

(Invalid Conditions)

For $x_1 \geq 6$

For $x_4 \geq 17$

Total allocated $= 6 + 17 = 23$

$r = 27 - 23 = 4$

$$C(5+4-1, 4) = C(8,4) = \frac{8!}{4!(8-4)!} = \frac{\overset{2}{\cancel{8}} \cdot 7 \cdot \cancel{6} \cdot 5 \cdot \cancel{4!}}{\cancel{4!}\ \cancel{4} \cdot \cancel{3} \cdot \cancel{2} \cdot 1}$$

$$= 2 \cdot 7 \cdot 5 = 70$$

Valid Solutions $=$ Total ways $-$ Invalid ways

$= 495 - 70$

$= 425$ Integer Solutions

5. Show that $\sum_{k=0}^{n} 2^k C(n,k) = 3^n$

use the Binomial Theorem

$$(x+y)^n = \sum_{k=0}^{n} C(n,k)\, x^k y^{n-k}$$

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let $x = 2$ and $y = 1$, then substitute :

$$(2+1)^n = \sum_{k=0}^{n} C(n,k)\,(2)^k\,(1)^{n-k}$$

the equation simplifies to:

$$(3)^n = \sum_{k=0}^{n} C(n,k)\,2^k$$

$$3^n = \sum_{k=0}^{n} 2^k\,C(n,k)$$

This completes the proof